

MERRITT, BC, CanadaWMO: **715570**Lat: **50.1125N**Lon: **120.7781W**Elev: **1939**StdP: **13.70**Time Zone: **-8.00 (NAP)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.5	(c) 4.6	(d) -7.6	(e) 3.9	(f) -0.8	(g) -2.3	(h) 5.2	(i) 6.5	(j) 25.6	(k) 25.8	(l) 22.8	(m) 26.5	(n) 3.9	(o) 70	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 20.3	(c) 86.5	(d) 61.8	(e) 82.3	(f) 60.7	(g) 78.4	(h) 60.5	(i) 68.4	(j) 74.1	(k) 66.4	(l) 71.5	(m) 64.7	(n) 70.2	(o) 7.2	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	104.7	70.8	64.6	98.0	69.1	62.8	91.9	67.0	33.9	73.9	32.2	70.5	30.7	69.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	17.9	15.4	-6.7	92.3	9.2	5.5	-13.3	96.3	-18.7	99.6	-23.8	102.7	-30.5	106.7
			-8.9	70.7	9.0	3.5	-15.4	73.2	-20.7	75.3	-25.7	77.3	-32.3	79.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	24.9	26.5	34.8	42.7	51.8	58.6	65.3	65.6	58.8	47.6	37.8	28.8
	DBStd	16.17	10.66	9.82	8.06	5.94	6.20	6.12	4.80	4.47	6.42	6.81	9.75	9.83
	HDD50	3360	777	659	472	232	51	4	0	1	7	127	373	657
	HDD65	7327	1242	1079	936	670	411	210	56	45	201	539	816	1122
	CDD50	1671	0	0	1	12	106	261	473	485	272	54	7	0
	CDD65	163	0	0	0	0	1	17	64	64	16	1	0	0
	CDH74	1956	0	0	0	9	96	262	775	666	147	1	0	0
	CDH80	618	0	0	0	1	19	82	268	212	36	0	0	0
(14) Wind	WSAvg	5.4	5.0	5.7	5.8	7.1	5.5	6.3	5.5	4.9	4.8	4.7	5.0	4.9
(15) Precipitation	PrecAvg	13.9	1.1	0.7	0.8	0.7	1.4	1.8	1.2	1.0	1.0	1.2	1.5	1.4
	PrecMax	18.4	2.1	1.3	1.4	1.5	3.1	3.2	3.5	2.4	2.3	2.2	2.9	3.4
	PrecMin	10.6	0.5	0.1	0.2	0.2	0.4	0.7	0.0	0.2	0.2	0.5	0.5	0.3
	PrecStd	2.1	0.4	0.3	0.3	0.3	0.7	0.7	0.7	0.6	0.6	0.5	0.6	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	50.1	60.8	72.8	82.5	88.1	92.7	91.8	85.0	70.1	59.5	52.9
		MCWB	48.4	43.7	46.7	54.7	59.3	62.2	61.8	61.8	61.3	57.4	53.8	48.3
	2%	DB	46.3	46.2	54.7	62.6	75.9	81.4	86.8	85.9	78.0	65.3	56.7	48.6
		MCWB	43.1	41.6	44.4	48.1	56.4	61.1	61.6	61.9	60.3	58.0	53.8	45.5
	5%	DB	42.5	43.2	50.5	57.3	70.0	75.4	82.3	81.0	73.2	61.9	53.7	45.9
		MCWB	39.3	38.6	41.8	45.7	53.9	58.7	60.8	60.8	60.2	55.0	50.8	42.5
	10%	DB	38.8	39.9	46.8	53.5	64.6	70.7	77.5	77.0	69.5	58.6	50.9	42.7
		MCWB	36.1	36.1	40.1	43.7	51.7	57.7	60.9	61.1	59.0	52.1	47.6	39.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	45.7	48.5	55.0	61.6	68.6	70.9	70.9	67.9	64.4	56.8	49.9
		MCDB	50.3	47.3	57.8	71.0	76.4	76.1	78.8	77.1	73.6	65.8	57.9	50.9
	2%	WB	44.0	42.5	45.5	49.9	57.5	64.9	68.2	68.7	65.1	59.6	53.7	46.5
		MCDB	45.4	45.9	52.7	58.8	73.0	72.6	74.2	73.6	69.9	62.7	55.4	47.6
	5%	WB	40.0	39.4	43.2	47.2	55.2	62.0	66.1	66.6	63.4	55.7	50.8	42.2
		MCDB	42.6	42.5	49.5	55.2	67.5	70.7	70.9	71.3	68.2	59.7	52.6	44.6
	10%	WB	36.9	36.5	41.1	45.0	53.3	59.9	64.0	64.8	61.6	52.8	47.5	38.9
		MCDB	39.4	40.1	46.7	52.0	63.6	69.2	70.4	70.3	66.8	57.0	49.8	41.8
(35) Mean Daily Temperature Range	MDBR		12.7	14.6	15.4	16.4	18.3	17.7	20.0	20.3	18.7	16.2	12.2	12.2
	5% DB	MCDBR	14.8	17.4	22.5	26.9	32.0	30.4	34.5	34.0	28.1	20.2	13.6	15.2
		MCWBR	13.0	13.3	13.3	14.5	14.6	13.2	12.2	13.5	14.4	13.3	11.2	14.1
	5% WB	MCDBR	14.9	16.6	20.3	23.5	26.6	21.6	16.0	17.0	17.3	14.4	12.3	16.0
		MCWBR	14.0	14.3	13.7	14.0	13.5	11.8	9.6	10.0	10.6	11.8	11.3	15.6
(40) Clear-Sky Solar Irradiance	taub		0.284	0.282	0.312	0.350	0.368	0.355	0.346	0.354	0.334	0.326	0.304	0.276
	taud		2.375	2.438	2.428	2.350	2.356	2.418	2.473	2.457	2.492	2.491	2.436	2.410
	Ebn at Noon		242	277	285	284	282	286	287	279	274	255	232	229
	Edh at Noon		20	24	30	36	37	35	33	32	28	23	19	17
(44) All-Sky Solar Radiation	RadAvg		323	616	990	1441	1722	1846	2056	1750	1219	707	343	247
	RadStd		34	50	79	112	142	169	166	111	102	73	36	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page